

F1 | Adaptive-prior architecture and operating hypotheses

A

Shared autoencoder

3,000 HVGs → encoder [256,128]
10-D latent representation
reconstruction objective shared
across all three variants

B

Adaptive DPMM prior

90% reconstruction warmup
Bayesian mixture refit
every 10 epochs
cluster count inferred online

C

Conditional flow refinement

transport toward DPMM centres
flow weight $\lambda_{FM} = 0.1$
noise scale = 0.5
projection-fidelity hypothesis

D

Pre-specified operating points

Pure-AE: label concordance
DPMM-Base: compact geometry
DPMM-FM: projection fidelity
No universal-best assumption